

Part A · Multiple Choice

2 points each

1. **a) wind speed, how long the wind blows, and the fetch**

pp. 227–228 — the fetch is the area of water the wind blows across. With 92 km/h winds blowing at least 69 hours, waves average nearly 15 m high.

2. **c) one-half of the wavelength**

p. 229 — this depth is the wave base. Because ocean wavelengths rarely exceed 200 m, the wave base only interacts with the bottom near shore, where it slows and steepens the waves until they break.

3. **d) the longshore current in the surf zone plus swash and backwash on the beach**

p. 231 — swash pushes sediment up the beach at an angle while backwash carries it straight back down, moving particles in a zigzag. Together with the longshore current this shifts enormous volumes of sediment along coasts.

4. **a) a spit**

p. 234 — a spit that extends far enough to close off a bay becomes a baymouth bar; a deposit connecting the mainland to a near-shore island is a tombolo.

5. **b) The U.S. Gulf Coast from Texas to Florida and the Atlantic Coast to Massachusetts**

p. 235 — they form where coastal sediment is abundant and relief is low. North of Boston the coast turns rocky, partly because of post-glacial crustal rebound.

6. **b) slowing the longshore current**

p. 240 — the trapped sediment is sediment that would otherwise have continued down the coast, which is why groin fields such as New Jersey's worsen erosion farther along the beach.

Part B · True or False

1 point each

7. **TRUE**

p. 232 — refraction concentrates energy on headlands while the intervening bays receive less, so headlands erode into caves, arches, stacks, and wave-cut platforms.

8. **FALSE**

It is the reverse. Rough winter seas with short-wavelength, high-amplitude waves strip sand offshore into sandbars; the gentler long-wavelength waves of summer push it back, making the beach wider and flatter. (p. 234)

Part C · Short Answer

3 points each

9. **Most beach sediment comes from rivers. A dam traps the river's sediment load in its reservoir, so for as long as the reservoir is filling, that sediment never reaches the sea and the beaches downcoast are starved.**

Rubric: 1 pt — most of the sediment forming coastal beaches is delivered by large rivers. 1 pt — when a river is dammed its sediment load settles out in the reservoir instead, and this continues for the century or two the reservoir takes to fill. 1 pt — without resupply, beaches within tens of kilometers of the river mouth (or more) erode, including spits, baymouth bars, and tombolos. Credit the conclusion the text draws: to take care of beaches, you have to take care of rivers. (p. 241)

10. Eustatic (global, from ice volume or seafloor shape), isostatic (local, from crustal uplift or subsidence), and tectonic (local, from plate processes).

Rubric: 1 pt each. **Eustatic** — global changes driven by the volume of glacial ice on land or by changes in seafloor shape from plate tectonics; roughly 125 m of rise over the past 20,000 years from glacial melting, now running about 3.2 mm/year since 1992. **Isostatic** — local changes from crust subsiding or rebounding, chiefly as ice is added or removed; several hundred meters of rebound around Hudson Bay since the last glaciation. **Tectonic** — local changes from plate processes, such as the roughly 1 mm/year uplift on western Vancouver Island caused by subduction of the Juan de Fuca Plate. (pp. 236–238)